Group II, Row IV

(checked; corrections marked in red)

Summary of corrections.

1. In the $\mathcal{P}_{gg}$ rewriting (eq. (1.15)) the prefactor must be $\frac{g^4}{16\pi^4}$, not $\frac{g^4 N_c}{16\pi^4}$: $\mathcal{P}_{gg}$ carries its own $2N_c$, so $-\frac{g^4 N_c}{8\pi^4}\hat{p} = -\frac{g^4}{16\pi^4}\mathcal{P}_{gg,+}$. The subsequent DGLAP/RC blocks (violet/green) were already written correctly, so only this one intermediate equation was off (by N_c).

2. The JIMWLK term in z-integral form (eq. (1.12)) must have a + sign: by eq. (1.11), $[-\frac{2}{\epsilon} - 2\gamma_E - \log] = +\frac{1}{\pi}\int_z K$, so no extra minus appears. (The analogous substitution for the DGLAP soft channel, eq. (1.19), was done correctly: there $[+\frac{2}{\epsilon} + \ldots] = -\frac{1}{\pi}\int_z K$ turned the − prefactor into +.) The sign is also fixed in the final sum and in the JIMWLK category.

3. In the final sum and in the "Finite terms" category, the first entry $(+\frac{g^4 N_c}{8\pi^5}\int\frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\int K(y,y',z)\,\mathcal{K}\cdot\mathcal{K}'\,W(y,y'))$ must be **deleted**: it is the DGLAP soft channel itself (eq. (1.19)), which is already contained in the DGLAP category through the full $\mathcal{P}_{gg}$. Keeping both counts that channel twice. What is added to the cross-section is only the exact zero = −(subtraction integral)+(its closed value); the remaining finite entries are the subtraction S, the boundary term, the endpoint term and the azimuthal term, with S + boundary + BFKL = 0 identically.

4. Minor: the $\delta(1-\xi)$ integration needs no $\vee\to\infty$ (the δ sits at $\xi=1$ inside $[\frac{k^+}{\vee},1]$, picking up the LO phase); and the LHS of the running-coupling category is relabeled $d^3N_{\rm RC}$.

1 The derivation

The setup for the cross section:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'}e^{-ik(w'-w)}\int_{y,y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+{+}k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z)$$
$$-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+{+}k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[\bar{\mathbb{C}}^{dac}_{1jik}(p^+{+}k^+,y-w;p^+,w-z)$$
$$U^{db}(z)U^{ce}(y)\mathbb{A}^{(1)\,e}_k(p^+{+}k^+,-y)-U^{db}(z)\bar{\mathbb{C}}^{dac}_{1jik}(p^+{+}k^+,y-w;p^+,w-z)\bar{\mathbb{A}}^{(1)\,c}_k(p^+{+}k^+,-y)\Big]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w',x'}e^{-ik(w'-w)}\int_{x,y,y'}\frac{(w-z)^m(x-y)^k(w'-z)^{m'}(x'-y')^{k'}}{(w-z)^2(x-y)^2(w'-z)^2(x'-y')^2}\frac{p^+k^+}{(p^++k^+)^4}$$
$$\delta^{(2)}\left[y-w+\frac{p^+}{(p^++k^+)}(w-z)\right]\times\delta^{(2)}\left[y'-w'+\frac{p^+}{(p^++k^+)}(w'-z)\right]\times$$
$$\left[\delta_{km}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{kj}\right]\times\left[\delta_{k'm'}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm'}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im'}\,\delta_{k'j}\right]$$
$$\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cda}\,U^{ce}(y)\,U^{db}(z)\,\rho^{e}(x)-f^{cda}\,U^{db}(z)\,U^{ce}(x)\,\rho^{e}(x)\right] \tag{1.1}$$

Integrating w and w',

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)\,\frac{(k^++p^+)}{k^+}}\int_{x,y,y',x'}\frac{(y-z)^m(x-y)^k(y'-z)^{m'}(x'-y')^{k'}}{(y-z)^2(x-y)^2(y'-z)^2(x'-y')^2}$$
$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^++p^+)^2}+\frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})+\frac{p^+}{k^{+2}}\,\delta_{mm'}\delta_{kk'}+\frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times$$
$$\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cda}\,U^{ce}(y)\,U^{db}(z)\,\rho^{e}(x)-f^{cda}\,U^{db}(z)\,U^{ce}(x)\,\rho^{e}(x)\right] \tag{1.2}$$

Reducing Wilson lines, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)}\,e^{-ik\,(y'-y)\,\frac{p^+}{k^+}}\int_{x,y,y',x'}\frac{(y-z)^m\,(x-y)^k\,(y'-z)^{m'}\,(x'-y')^{k'}}{(y-z)^2\,(x-y)^2\,(y'-z)^2\,(x'-y')^2}$$
$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^++p^+)^2}+\frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})+\frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'}+\frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.3}$$

We are performing the z integral and using $\overline{MS}$ and $d_\perp = 2-\epsilon$, we get

$$\int_z\frac{(y-z)^m\,(y'-z)^{m'}}{(y-z)^2\,(y'-z)^2} = -\frac{\pi}{2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\delta^{mm'}\,-\pi\,\frac{(y'-y)^m\,(y'-y)^{m'}}{(y'-y)^2} \tag{1.4}$$

This gives us

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}$$
$$\left[\frac{p^+}{(k^++p^+)^2}+\frac{p^+}{k^{+\,2}}+\frac{1}{p^+}\right]\times\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\,\frac{(x-y)^k\,(x'-y')^{k'}\,(y'-y)^m\,(y'-y)^{m'}}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}$$
$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^++p^+)^2}+\frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})+\frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'}+\frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.5}$$

Simplifying (the antisymmetric $\frac{2}{k^+}$ channel drops: contracted with the symmetric $\delta^{mm'}$ or $\frac{r^m r^{m'}}{r^2}$ it vanishes), we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{p^+}{(k^++p^+)^2}+\frac{p^+}{k^{+\,2}}+\frac{1}{p^+}\right]\times$$
$$\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\,\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,\frac{d_\perp\,p^+}{(k^++p^+)^2}\right.$$
$$\left.+\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[\frac{p^+}{k^{+\,2}}+\frac{1}{p^+}\right]\right]\times\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.6}$$

Using $\xi = \frac{k^+}{(p^++k^+)}$, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^++\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\frac{1}{\xi^2}$$
$$\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\,\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$+\,\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^++\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\xi^2}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\,\xi(1-\xi)\right.$$
$$\left.+\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\right]\times\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.7}$$

The cross section is given by

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\times\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]$$
$$\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\right]\right]\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\left[\log\frac{(\vee-k^+)\,k^+}{\vee\,\Lambda}\right]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.$$
$$\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+-\Lambda}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}$$
$$\times\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.8}$$

Using $d_\perp = 2-\epsilon$, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\right]$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\left[\log\frac{k^+}{\Lambda}+\log\left(1-\frac{k^+}{\vee}\right)\right]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.9}$$

The JIMWLK term is straightforward and is given by

$$\frac{d^3N_{\rm JIMWLK}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[-\frac{2}{\epsilon}-2\gamma_E-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.10}$$

This term can be written as using the following relation

$$\left[-\frac{2}{\epsilon}-2\gamma_E-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]=\frac{1}{\pi}\int_z\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2} \tag{1.11}$$

$$\frac{d^3N_{\rm JIMWLK}}{d^3k} = +\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,z,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.12}$$

The sign here is $+$, not $-$: the bracket of (1.10) is $[-\frac{2}{\epsilon}-2\gamma_E-\log] = +\frac{1}{\pi}\int_z K(y,y',z)$ by (1.11), a straight substitution with no extra minus sign. Consistency check: the same substitution is made below for the DGLAP soft channel, going to (1.19) — there the bracket is $[+\frac{2}{\epsilon}+\dots] = -\frac{1}{\pi}\int_z K$, and the $-$ prefactor correctly became $+$. Done the same way here, the $+$ prefactor of (1.10) stays $+$.

The remaining term is as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\right]$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.13}$$

Now, let us move on to extract the DGLAP and running-coupling contribution. In the above equation, we denote

$$\mathcal{P}_{gg,+}(\xi) = 2N_c\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\right]$$

and thus we write the gluon-gluon splitting function ($\mathcal{P}_{gg}$) as follows:

$$\mathcal{P}_{gg}(\xi) = \mathcal{P}_{gg,+}(\xi)+\frac{\beta_0}{2}\,\delta(1-\xi) \tag{1.14}$$

So, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^4}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\mathcal{P}_{gg}(\xi)-\frac{\beta_0}{2}\,\delta(1-\xi)\right]$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.15}$$

The prefactor of the first term is $\frac{g^4}{16\pi^4}$, without N_c: $\mathcal{P}_{gg,+}$ carries its own $2N_c$, so $-\frac{g^4N_c}{8\pi^4}\,\hat{p}_+(\xi) = -\frac{g^4N_c}{8\pi^4}\cdot\frac{\mathcal{P}_{gg,+}}{2N_c} = -\frac{g^4}{16\pi^4}\,\mathcal{P}_{gg,+}$. (With $\frac{g^4N_c}{16\pi^4}$ the DGLAP and RC terms below would come out N_c times too large; as a check, the $\frac{\beta_0}{2}\delta(1-\xi)$ piece must give exactly the RC term $+\frac{g^2\beta_0}{16\pi^2}\,L_\epsilon\,\frac{d^3N_{\rm LO}}{d^3k}$, which it now does.)

In the first term, integrating with respect to ξ for the β_0 term (the $\delta(1-\xi)$ sits at $\xi=1$ inside $[\frac{k^+}{\vee},1]$: the factor $\frac{1}{\xi^2}$ becomes 1 and the phase becomes the LO phase; no limit $\vee\to\infty$ is needed), we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2}{8\pi^2}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2\,\beta_0}{16\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.16}$$

violet - DGLAP and green - running coupling. The remaining term is as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.17}$$

Now, let us move on to obtaining the BFKL term. The DGLAP term in eq. (1.16) contains a term which is divergent when $\xi\to 0$. That term is

$$D_{\rm soft} = -\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.18}$$

(we call it $D_{\rm soft}$; it stays *inside* the DGLAP term, we only analyse it here). Rewriting it using (1.11), we get the following

$$D_{\rm soft} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.19}$$

This term has two features: one is the DGLAP part and the other is a hidden $\log\frac{\vee}{k^+}$ (when $\xi\to 0$). Using eq. (1.1),

$$y-w+(1-\xi)\,(w-z) = 0 = y'-w'+(1-\xi)\,(w'-z)$$
$$y-z-\xi(w-z) = 0 = y'-z-\xi(w'-z)$$

and setting $\xi\to 0$, we get

$$y-z=0=y'-z \implies y=y'=z$$

Implementing this in eq. (1.19), we get

$$\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)\to\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\left[U^{ce'}(z)-U^{ce'}(x')\right]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.20}$$

The phase transforms:

$$e^{-ik\cdot(y'-y)\frac{1}{\xi}}\to e^{-ik\cdot(w'-w)}$$

Thus, we add and subtract this divergent piece with (1.19):

$$D_{\rm soft} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\Bigg\{$$
$$\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$-\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\left[U^{ce'}(z)-U^{ce'}(x')\right]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)\Bigg\}$$
$$+\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\left[\log\left(\frac{\vee}{k^+}\right)-\left(1-\frac{k^+}{\vee}\right)\right]\int_{x,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\left[U^{ce'}(z)-U^{ce'}(x')\right]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.21}$$

The BFKL term is:

$$\frac{d^3N_{\rm BFKL}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\left[U^{ce'}(z)-U^{ce'}(x')\right]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.22}$$

Assembling the full cross-section: what is added is only a zero. Since the DGLAP category below keeps the *full* $\mathcal{P}_{gg}$, the soft channel $D_{\rm soft}$ stays inside it and must **not** be listed again. What is inserted into the cross-section is only the exact zero

$$0 = \underbrace{-\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\int_{x,y,x',y',z} e^{-ik(y'-y)\frac{1}{\xi}}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2(y'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\,W(z,z)}_{\text{subtraction integral}} + \underbrace{\left[\log\tfrac{\vee}{k^+}-\left(1-\tfrac{k^+}{\vee}\right)\right]\times(\text{its c}}_{\text{BFKL + boundary}}$$

where $W(z,z) = [U^{ce'}(z) - U^{ce'}(x')][U^{ce}(z) - U^{ce}(x)]\rho^{e'}(x')\rho^{e}(x)$. The first term of the original final sum ($+\frac{g^4 N_c}{8\pi^5}\int \frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\int K(y,y',z)$ was $D_{\rm soft}$ itself (eq. (1.19)) and has been **deleted**: keeping it together with the full $\mathcal{P}_{gg}$ DGLAP term would count the soft channel twice. The role of the subtraction integral in the finite block is that, combined with $D_{\rm soft}$ inside DGLAP, the pair (1.21) (first two lines) is free of $\frac{2}{\epsilon}$, μ and of $\log\frac{\vee}{k^+}$.

We write the cross-section in different categories:

1.1 JIMWLK

$$\frac{d^3N_{\rm JIMWLK}}{d^3k} = + \frac{1}{(2\pi)^3}\frac{g^4 N_c}{8\pi^5}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,z,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\Big[U^{ce'}(y') - U^{ce'}(x')\Big]\,[U^{ce}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x) \tag{1.23}$$

(sign corrected, see (1.12)).

1.2 BFKL

$$\frac{d^3N_{\rm BFKL}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\Big(\frac{\vee}{k^+}\Big)\int_{x,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\Big[U^{ce'}(z) - U^{ce'}(x')\Big]\,[U^{ce}(z) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x) \tag{1.24}$$

1.3 DGLAP

$$\frac{d^3N_{\rm DGLAP}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2}{8\pi^2}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon} + 2\,\gamma_E + \log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times$$
$$\Big[U^{ce'}(y') - U^{ce'}(x')\Big]\times[U^{ce}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x) \tag{1.25}$$

1.4 Running coupling

$$\frac{d^3N_{\rm RC}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2\,\beta_0}{16\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon} + 2\,\gamma_E + \log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times$$
$$\Big[U^{ce'}(y') - U^{ce'}(x')\Big]\times[U^{ce}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x) \tag{1.26}$$

1.5 Finite terms

(the former first entry — the soft channel $D_{\rm soft}$ in z-form — has been removed: it is already inside the DGLAP term above)

$$\frac{d^3N_{\rm Finite}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\Big[U^{ce'}(z) - U^{ce'}(x')\Big]\,[U^{ce}(z) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\Big[U^{ce'}(z) - U^{ce'}(x')\Big]\,[U^{ce}(z) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\log\Big(1-\frac{k^+}{\vee}\Big)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big]$$
$$\Big[U^{ce'}(y') - U^{ce'}(x')\Big]\,[U^{ce}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - 2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\Big[U^{ce'}(y') - U^{ce'}(x')\Big]\,[U^{ce}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x) \tag{1.27}$$

Bookkeeping. (i) (first entry, the subtraction S) + (second entry, the boundary term) + (BFKL category) = 0 identically, so the five categories sum back to (1.9) exactly. (ii) Per ξ, S pairs with the soft channel of the DGLAP category into a combination free of $\frac{2}{\epsilon}$, γ_E, μ and of $\log\frac{\vee}{k^+}$ (the curly bracket of (1.21)). (iii) The third entry ($\log(1-\frac{k^+}{\vee})$) is the plus-prescription convention term: with the standard $[0,1]$ plus distribution in $\mathcal{P}_{gg}$ it is absorbed into the DGLAP term. Neither it nor the azimuthal term contains a rapidity logarithm.